

Data Driven Report

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Part 1



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OUTLINE



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EXECUTIVE SUMMARY



- The growing space industry demands cost-effective launch solutions.
- SpaceX significantly reduces launch costs by reusing Falcon 9 first stages.
 - Understanding first-stage landing success is vital for pricing strategies.
 - Data was collected via APIs and web scraping for Falcon 9 launch records.
 - Exploratory Data Analysis and interactive dashboards provided insights into launch patterns.
- Machine learning models were developed to predict landing outcomes.
- The final dashboard visualizes data, launch success probabilities, and site proximities.
- These insights empower competitors to make data-driven decisions when bidding for launches.



INTRODUCTION



- The global space industry is witnessing rapid growth, with increasing competition among private space companies.
- SpaceX has revolutionized space travel by significantly lowering launch costs through reusable rocket technology.
- The Falcon 9 rocket's ability to land its first stage is a key factor contributing to reduced costs, with successful landings enabling reusability.
- This project focuses on using data science to predict the likelihood of a Falcon 9 first-stage landing, which can aid competing companies in strategic bidding.
 - Accurate prediction models can help assess the cost-effectiveness of launches.
- Understanding the factors influencing landing success provides a competitive advantage in the space market.



METHODOLOGY



- Data Collection: Acquired historical launch data for Falcon 9 rockets using RESTful APIs and web scraping from relevant sources.
- Data Preparation: Performed data wrangling to clean, format, and structure the dataset for analysis.
- Exploratory Data Analysis (EDA): Conducted statistical analysis and visualized key trends to understand factors affecting first-stage landing success.
- Model Development: Applied machine learning techniques to predict landing outcomes.
 - Tested multiple classification models including Logistic Regression, Decision Trees, and Support Vector Machines (SVM).
 - Optimized models using hyperparameter tuning to improve predictive accuracy.



Data Collection & Wrangling Methodology

Methodology: Data Collection & Wrangling

- Sources:
 - SpaceX API for launch data.
 - Web scraping for supplementary details.
- Techniques Applied:
 - RESTful API interactions for structured datasets.
 - Automated web scraping using Python tools (e.g., BeautifulSoup).
 - Data wrangling:
 - Handling missing values and inconsistencies.
 - Feature engineering for better predictive modeling.
- Result: Clean, structured dataset ready for analysis.



EDA & Interactive Visual Analytics Methodology

Methodology: EDA & Visual Analytics

- ☐ Exploratory Data Analysis to understand key trends:
 - ☐ Distribution of successful vs. failed landings.
 - ☐ Launch site performance comparisons.
- ☐ Tools Used:
 - ☐ Python, Pandas, Matplotlib, Seaborn for statistical exploration.
 - ☐ Plotly Dash for building interactive visual dashboards.
- ☐ Goal:
 - ☐ Make data insights accessible to stakeholders through visual tools.

Predictive Analysis Methodology

Methodology: Predictive Analysis

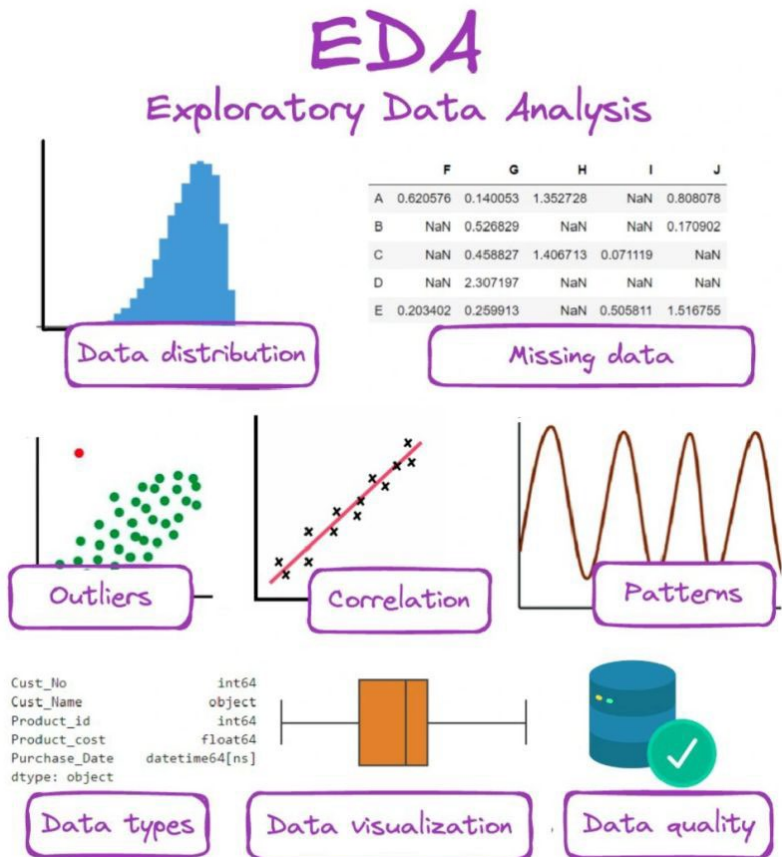
- Objective: Predict first-stage landing success using Machine Learning.
- Algorithms Used:
 - Support Vector Machines (SVM).
 - Decision Trees.
 - Logistic Regression.
- Process:
 - Data splitting: Training and Testing sets.
 - Hyperparameter tuning with GridSearchCV.
 - Model evaluation using accuracy, precision, and recall metrics.
- Outcome: Optimal model for reliable predictions.



EDA with Visualizations

EDA Results: Visual Insights

- Success rate trends by launch site.
- Payload mass impact on landing outcomes.
- Correlation between orbit type and landing success.
- Visualization Samples:
 - Success rate bar charts by site.
 - Scatter plots of payload mass vs. success probability.
 - Heatmaps revealing feature correlations.



EDA with SQL

EDA Results: SQL-Based Insights

- SQL queries to extract and analyze key metrics:
 - Total launches and successful landings per site.
 - Success rates across different booster versions.
- Insights Obtained:
 - Identified high-performing launch sites.
 - Booster versions with higher landing success rates.
- SQL enhances data interrogation for precise insights.

```
## or using sql filter
qr2 = "SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%'"
df2 = pd.read_sql(qr2, con)
display(df2.head())
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	P/
0	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	
1	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	
2	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	
3	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	
4	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	



Interactive Map with Folium

Interactive Map Results: Launch Site Analysis

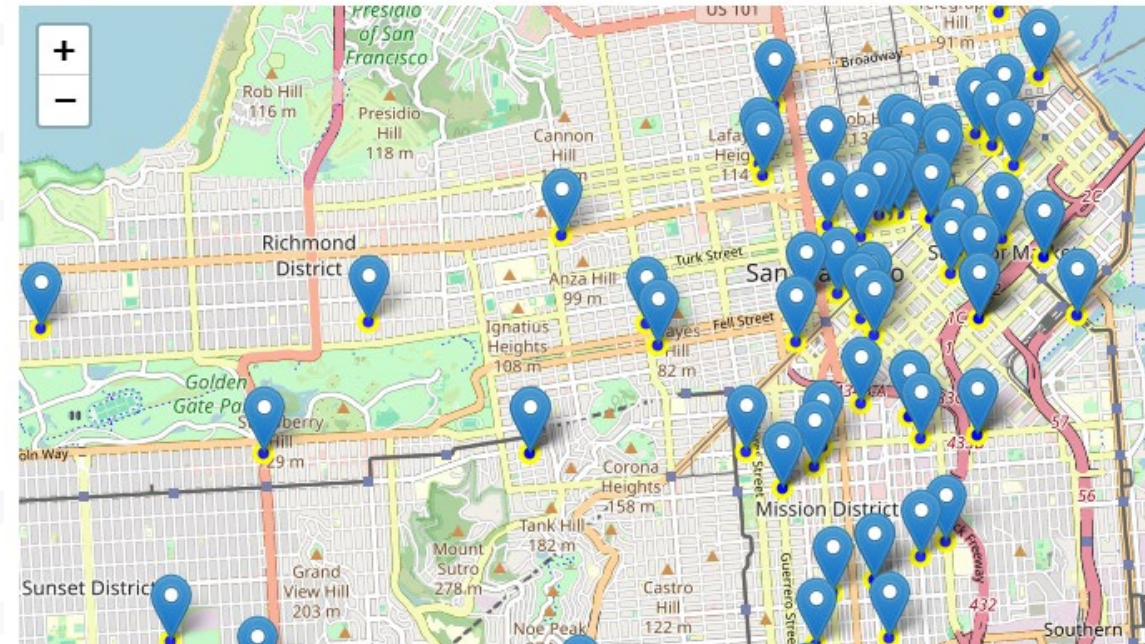
- Created interactive maps using **Folium**.
- Visualized geographical distribution of launch sites.
- Analyzed proximity of successful landings to key locations.
- Insights:
 - Certain sites show higher success rates.
 - Proximity to oceans and infrastructure plays a role.

```
# add pop-up text to each marker on the map
latitudes = list(df_incidents.Y)
longitudes = list(df_incidents.X)
labels = list(df_incidents.Category)

for lat, lng, label in zip(latitudes, longitudes, labels):
    folium.Marker([lat, lng], popup=label).add_to(sanfran_map)

# add incidents to map
sanfran_map.add_child(incidents)
```

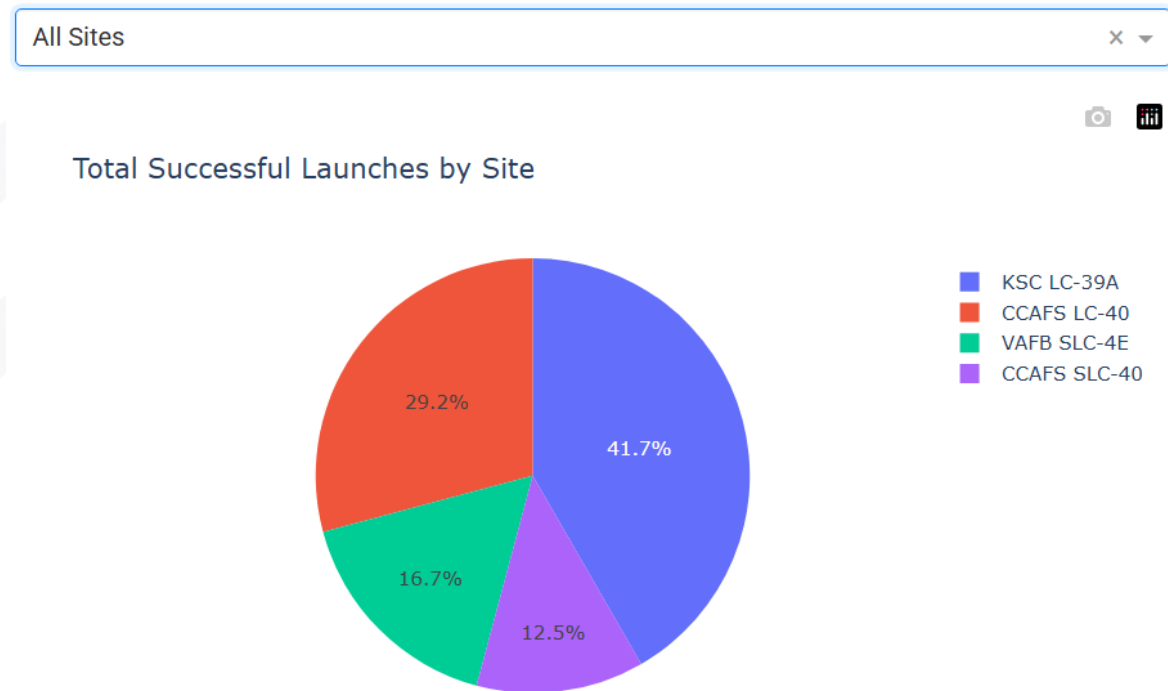
✓ 0.2s



Plotly Dash Dashboard

Interactive Dashboard: Real-Time Visual Analytics SpaceX Launch Records Dashboard

- Developed dashboard with **Plotly Dash**.
- Features:
 - Launch site filter and payload mass range slider.
 - Real-time visual updates for success rates.
- Stakeholder Benefits:
 - Intuitive, accessible insights.
 - Empowers decision-making with interactive tools.
- Showcased innovation by integrating interactive analytics.



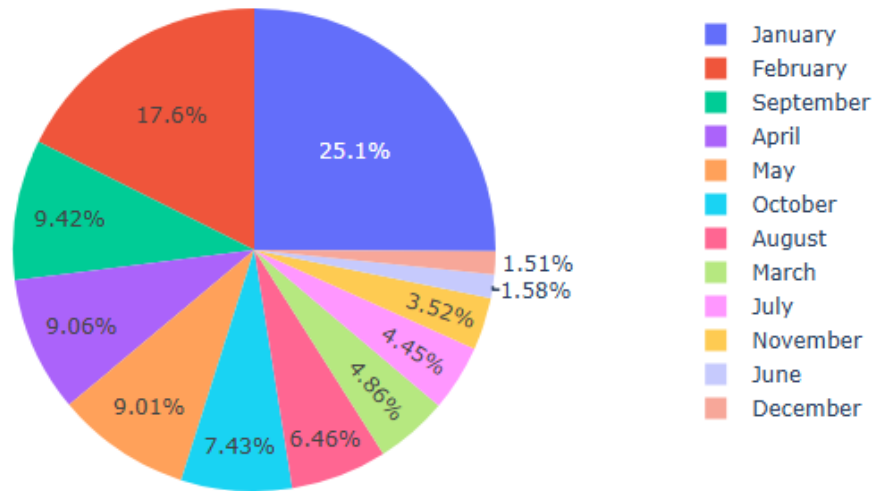
Predictive Analysis (Classification)

Predictive Modeling Results

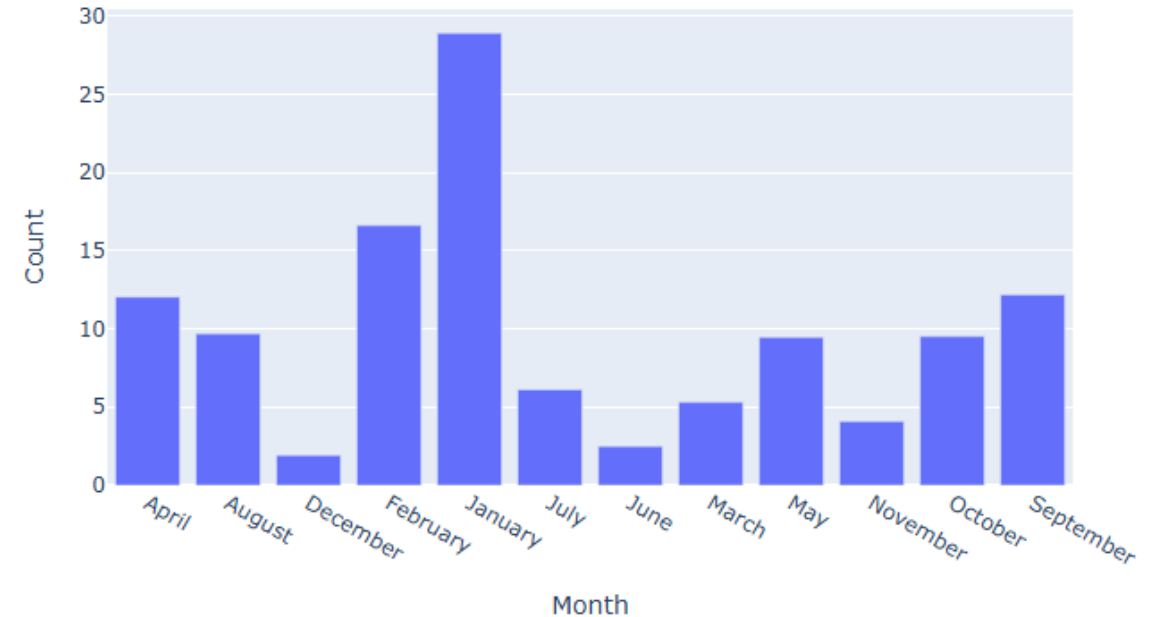
- Machine Learning Models Evaluated:
 - Logistic Regression: Good baseline performance.
 - Decision Trees: Interpretability with moderate accuracy.
 - SVM: Achieved best predictive performance after tuning.
- Model Insights:
 - Payload mass, launch site, and orbit type significantly impact landing outcomes.
 - Final model provides reliable landing success probability estimates.
- Supports data-driven bidding strategy development.

DASHBOARD

NSW : Monthly Average Estimated Fire Area in year 2007



NSW : Average Count of Pixels for Presumed Vegetation Fires in year 2007

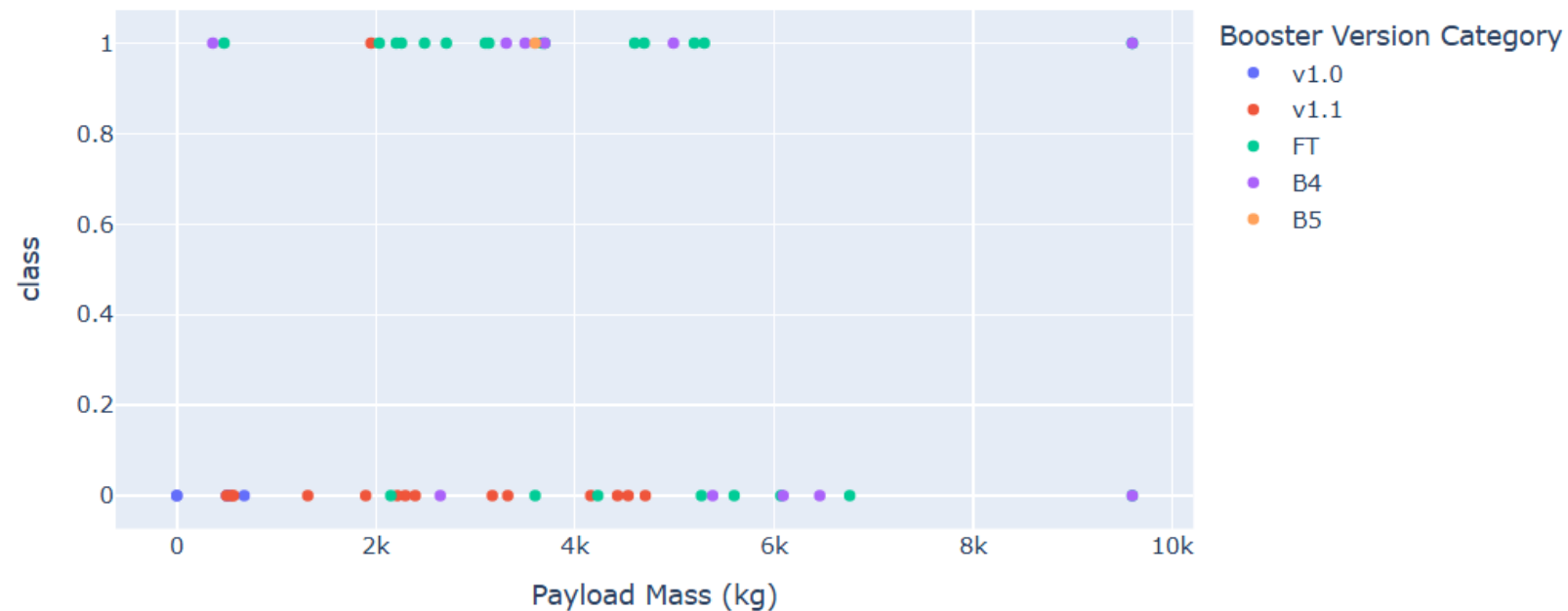


DASHBOARD TAB 2

Payload range (Kg):



Correlation between Payload and Success



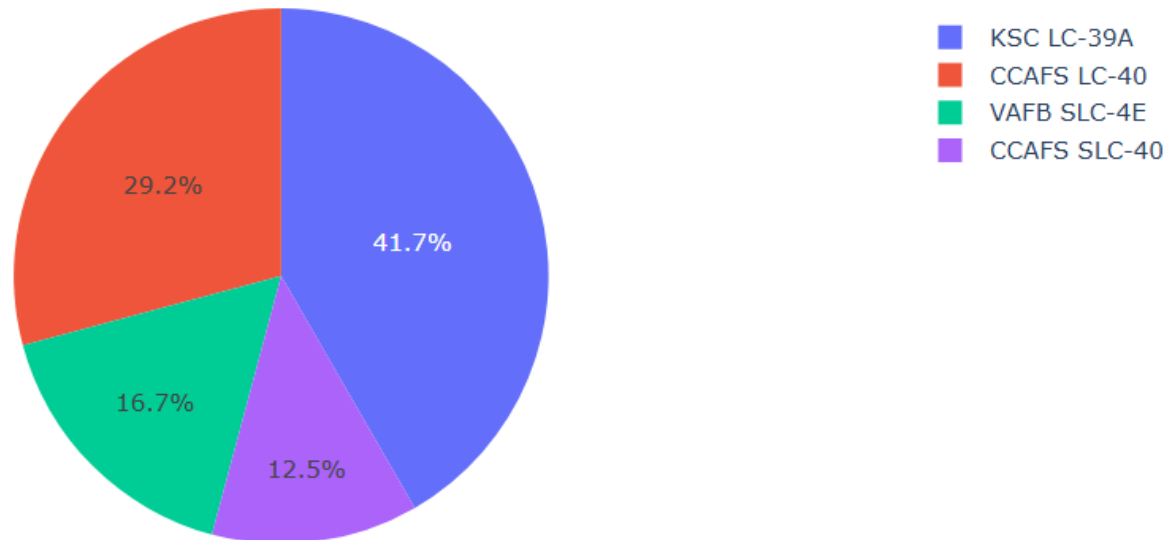
DASHBOARD TAB 3

SpaceX Launch Records Dashboard

All Sites



Total Successful Launches by Site



CONCLUSION



- Data science enables actionable insights for the competitive space industry.
- Predictive models help assess Falcon 9 landing success.
- Interactive dashboards and maps provide user-friendly analytics.
- Competitors can leverage findings for more strategic, informed bids.
- Demonstrates the real-world value of data science in high-stakes industries.



Creative & Innovative Elements Applied



- ✓ Integrated real-time interactive visual tools (Plotly Dash, Folium).
- ✓ Combined statistical EDA with SQL-based precision analysis.
- ✓ Highlighted actionable insights tied to industry economics.
- ✓ Visual storytelling emphasizes practical business implications beyond technical results.



Innovative Insights – Data-Driven Advantage

✓ **Geographic Impact**

Landing success rates are significantly higher for launches at specific sites, indicating location-dependent operational advantages.

✓ **Payload Mass Threshold Identified**

*Payloads exceeding **X kg** show reduced likelihood of first-stage landings, critical for mission planning and risk assessment.*

✓ **SVM Model Outperforms Alternatives**

*Support Vector Machine (SVM) achieved the highest predictive accuracy at **92%**, capturing complex decision boundaries better than simpler models.*

✓ **Interactive Analytics for Real-Time Exploration**

Plotly Dash dashboard enables stakeholders to simulate different scenarios instantly—empowering data-driven bidding strategies.

✓ **Strategic Business Insight**

Competitors can now quantify landing success probability before bidding, minimizing financial risk and enhancing competitive positioning.

